

Substrate-Clifford Dimensional Identity v0.1

Author: Lyra Date: 2026-06-01 (Monday) ~14:46 EDT Status: v0.1 SYNTHESIS — substantive substrate-primary algebraic identity Trigger: Elie Toy 3703 PASS 5/5 substantive observation: $\dim \text{Cl}(5,2) = 2^g = 128$ substrate primary ★

1. Statement of the Identity

The substrate signature is (5, 2) (matching the indefinite-orthogonal group $\text{SO}_0(5,2) \supset K = \text{SO}(5) \times \text{SO}(2)$). The Clifford algebra of a real quadratic space of signature (p, q) has real dimension

$$\dim_{\mathbb{R}} \text{Cl}(p, q) = 2^{p+q}.$$

For the substrate signature:

$$\dim_{\mathbb{R}} \text{Cl}(5, 2) = 2^{5+2} = 2^7 = \boxed{128}.$$

Equivalently in substrate primaries:

$$\dim \text{Cl}_{\text{substrate}} = 2^g$$

where $g = 7$ is the substrate primary (dual Coxeter / multiplicity-7). This is a NEW substrate-primary algebraic identity:

$$g = \log_2 \dim \text{Cl}_{\text{substrate}}.$$

2. Why This Matters

Three structural observations:

(a) Substrate-primary closure of Clifford dimension. Until now, $g = 7$ has appeared in the substrate primary list as (i) the dual Coxeter number of $B_2 + 1$, (ii) the multiplicity in the Heckman-Opdam ladder, and (iii) one of the five substrate primaries with mass-relation reach. The Clifford identity gives g a fourth substrate-natural role: it determines the dimension of the substrate Clifford algebra by the simplest possible relation (logarithm).

(b) Dirac restriction to 4D substrate slice. Casey-named #14 ($\text{codim } 4 \subset D_{IV}^5 = C_2 = 6$) gives the substrate-natural 4D Minkowski slice. The Clifford algebra of 4D Minkowski signature (3, 1) has

$$\dim \text{Cl}(3, 1) = 2^4 = 16.$$

This is the standard dimension of Dirac γ -matrix algebra (4×4 matrices = 16 real components). The substrate restriction $\text{Cl}(5, 2) \rightarrow \text{Cl}(3, 1)$ is a substrate-natural embedding of substrate Clifford algebra into 4D physical Clifford algebra with codimension matching Casey #14.

(c) Connection to Substrate-Dirac (Lyra v0.1 + Elie Toy 3703). Both Lyra Substrate-Dirac derivation v0.1 and Elie Toy 3703 build the Dirac equation on the $V_{-}(1/2, 1/2)$ spinor K -type of D_{IV}^5 . The substrate-natural Clifford algebra $\text{Cl}(5, 2)$ of dimension 128 is the ambient algebra in which the substrate γ -matrices live; restriction to the 4D Minkowski slice gives standard Dirac γ -matrices in $\text{Cl}(3, 1)$.

3. Substrate-Primary Factorization Tracking

The full set of substrate-natural identities involving $g = 7$ now reads:

Identity	Form	Source
$g = \text{dual Coxeter } (B_2 + 1)$	$g = h^{\vee}(B_2) + 1$	Standard Lie theory
$g = \text{Heckman-Opdam multiplicity}$	$g = 2k + 1$ where $k=3$	FK Ch. XII-XIII
$g = \text{mass-ladder exponent}$	$(m_u + m_d)/m_e \approx 2^g/(N_c \cdot \pi)$	Grace INV-5451 substrate-clean form
$g = \log_2 \dim \text{Cl}(5,2) [\text{NEW}]$	$g = \log_2(128)$	This note + Elie Toy 3703

Per Cal #27 STANDING (cross-track convergence honestly framed): these are FOUR distinct algebraic roles for the SAME primary $g = 7$, NOT four independent confirmations.

4. What This Does NOT Claim

Per Cal #34 STANDING + Cal #35 STANDING (Two-Tier + Independence-Taxonomy):

(a) Tier 1 EXACT — the $\dim \text{Cl}(p,q) = 2^{(p+q)}$ identity is a textbook algebraic fact; the substrate-primary identification $g = \log_2 \dim \text{Cl}_{\text{substrate}}$ is exact in substrate primary arithmetic.

(b) NOT a new independent derivation leg. Per Cal #35-honest framing: this is ONE substrate-natural algebraic role for g , not a NEW independent route forcing D_{IV}^5 . The Strong-Uniqueness 10 legs (v1.5 STANDALONE) remain the operative count; this identity REFINES the substrate-primary algebraic taxonomy but does NOT add an 11th leg.

(c) Multi-week task: explicit construction of γ^μ as elements of $\text{Cl}(3, 1) \subset \text{Cl}(5, 2)$ substrate restriction with explicit substrate matrix forms ($\text{Cl}(5,2) \rightarrow \text{Cl}(3,1)$ embedding map). This is part of the Substrate-Dirac multi-week roadmap (see Lyra Substrate-Dirac Derivation v0.1).

5. Cross-References

- Lyra Substrate-Dirac Derivation v0.1 (Mon ~14:35 EDT)
- Elie Toy 3703 PASS 5/5 (Mon ~14:40 EDT) — original observation
- Casey-named #14 codim $4 \subset D_{IV}^5 = C_2$ (Monday v0.1)
- Strong-Uniqueness v1.5 STANDALONE (Monday) — 10 legs unchanged
- Grace INV-5451 $(m_u + m_d)/m_e \approx 2^g/(N_c \cdot \pi)$ substrate-clean form (Saturday)

6. Closure

The substrate-primary identity $g = \log_2 \dim \text{Cl}(5, 2) = \log_2 128 = 7$ adds a fourth substrate-natural algebraic role for g . The restriction $\text{Cl}(5, 2) \rightarrow \text{Cl}(3, 1)$ via Casey #14 codim 4 reproduces the standard Dirac γ -matrix dimension 16. Per Cal #35-honest: ONE substrate primary with FOUR substrate-natural roles, NOT four independent confirmations.

— Lyra, Mon 2026-06-01 ~14:46 EDT. Synthesis of Elie Toy 3703 substrate-primary observation. Substrate-Clifford dimensional identity v0.1.